

RunZone: System Architecture & Interview Guide

Full Stack Geospatial SaaS · PostGIS · ACWR Sports Science · LLM Guardrails · APScheduler

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1. Executive Overview & 60-Second Elevator Pitch

RunZone is an enterprise-grade endurance platform that transforms outdoor GPS running into competitive geospatial territory conquest while actively preventing soft-tissue overtraining injuries using the Acute:Chronic Workload Ratio (ACWR) mathematical model. The platform integrates real-time PostGIS polygon buffering, an automated sector decay engine, Groq/Gemini LLM ZoneCoach with clinical safety guardrails, Strava webhook ingestion, and live WebSocket telemetry.

Core Value Proposition (Why It Stands Out in Senior Engineering Interviews)

Spatial Computation: Performs server-authoritative 40-meter corridor dilation using PostGIS EPSG:3857 planar projection and GiST spatial indexing for sub-100ms collision detection.

Sports Medicine Engine: Implements Dr. Tim Gabbett's ACWR model to mathematically flag soft-tissue fatigue spikes before injury occurs.

Living Game Economy: Automated APScheduler daily sweeps degrade unrun sectors (-15 pts/day), releasing inactive zones to neutral.

Truth-in-UI Standards: Zero mocked or synthetic data fallbacks — honest empty states and full server-side deterministic algorithms.

2. High-Level System Architecture & Technology Stack

RunZone is architected as an asynchronous, decoupled multi-service system designed for high I/O throughput, low-latency geospatial querying, and robust background event processing.

Technical Stack Breakdown

Backend API: Python 3.11, FastAPI (Async ASGI), Pydantic v2 validation, Uvicorn worker pool.

Database & Spatial Engine: PostgreSQL 16 with PostGIS 3.4 (GeoAlchemy2, Shapely, GiST spatial indexing on Polygon corridors).

Task Scheduling: APScheduler 3.10 (AsyncIOScheduler) running in-process daily decay jobs with isolated DB sessions.

AI & Voice Engine: Groq Llama 3.3 70B Versatile, Whisper Large-v3 Turbo audio transcription, Medical Guardrails Engine.

Real-Time Telemetry: Native WebSockets Manager for instantaneous broadcast of conquests, decays, and splits.

Frontend App: React 19, TypeScript, Tailwind CSS, Leaflet/MapLibre 3D, Recharts dual-axis load curves.

Module Deep-Dives: Spatial Computation & Sports Science

Deep technical rationale, mathematical formulas, and algorithmic implementation

3. PostGIS Geospatial Territory Engine (Deep-Dive)

The territory conquest engine turns continuous GPS coordinates into real-world polygonal corridors. Unlike client-side approximations that are vulnerable to tampering, RunZone enforces strict server-side PostGIS geometric operations.

- **Coordinate Transformation & Planar Buffering:** Standard WGS84 coordinates (EPSG:4326 in degrees) are projected into Web Mercator (EPSG:3857 in planar meters). The server applies `ST_Buffer(geom_3857, 40.0)` to generate a precise 40-meter dilated running corridor around the athlete's track.
- **Spatial Overlap & Conquest Resolution:** When a corridor is saved, PostGIS executes an `ST_Intersects` query against existing territory zones. GiST (Generalized Search Tree) spatial bounding-box indexing reduces spatial overlap detection time from $O(N)$ to $O(\log N)$, completing in under 80ms even with dense urban zones.
- **Contested Sector Calculation:** If an overlap occurs, `ST_Difference` and `ST_Area` compute the newly stolen territory area in km^2 , updates the sector defense points, and dispatches a WebSocket broadcast payload to all active runners on that map sector.

4. Acute:Chronic Workload Ratio (ACWR) Injury Prevention Engine

The sports science engine is based on peer-reviewed research by Dr. Tim Gabbett (British Journal of Sports Medicine). It tracks how fast an athlete ramps up training volume relative to their historical chronic aerobic baseline.

ACWR Formula & Risk Zone Matrix

Acute Load (7-Day Fatigue): Sum of daily Training Impulse (TRIMP) or Session-RPE workload over the past 7 days: $\text{Acute} = \text{SUM}(\text{Workload}[t-6 \dots t])$.

Chronic Load (28-Day Fitness): Rolling daily average workload over the past 28 days: $\text{Chronic} = (\text{SUM}(\text{Workload}[t-27 \dots t])) / 4$.

ACWR Calculation: $\text{ACWR} = \text{Acute Load} / \text{Chronic Load}$

0.80 - 1.30 (Sweet Spot): Optimal training zone. Lowest risk of soft-tissue and ligament injury (~10-15%).

1.30 - 1.50 (Overreaching): Elevated fatigue zone. Risk of injury doubles (~30-50%). Coach advises low Zone 2 recovery.

> 1.50 (Danger Zone): High injury hazard zone. Soft-tissue breakdown risk spikes 2x-4x. Rest day strictly enforced.

5. Autonomic Readiness & HRV Recovery Hub

RunZone computes a daily morning Readiness Score (0–100%) from Heart Rate Variability (rMSSD in ms), Resting Heart Rate (bpm), and Sleep quality/duration. Baseline deviations are normalized into a weighted composite metric:

- **HRV rMSSD Deviation Score:** $S_{\text{HRV}} = \text{clamp}(0, 100, 50 + ((\text{HRV} - 64) / 15) * 25)$ — captures parasympathetic autonomic recovery.
- **Resting HR Deviation Score:** $S_{\text{RHR}} = \text{clamp}(0, 100, 50 + ((52 - \text{RHR}) / 5) * 25)$ — captures cardiovascular stress and elevated basal strain.
- **Sleep Score & Composite:** $S_{\text{Sleep}} = (\text{sleep_hours} / 8.0) * \text{sleep_quality}$. Final Score = $\text{round}(0.40 * S_{\text{HRV}} + 0.30 * S_{\text{RHR}} + 0.30 * S_{\text{Sleep}})$.

Automated Jobs, AI Safety Guardrails & Enterprise Security

Scheduled background processing, LLM guardrail architecture, and OAuth/JWT flows

6. Sector Decay Engine & In-Process APScheduler

To maintain competitive dynamic gameplay without requiring human intervention, RunZone runs an automated sector decay engine registered inside FastAPI's lifespan context manager.

Decay Engine Workflow & Mechanics

Scheduler Setup: AsyncIOScheduler starts during FastAPI lifespan startup, executing daily at 03:00 UTC (CronTrigger).

Isolated DB Session: The scheduled worker opens its own AsyncSessionLocal context, decoupling it from HTTP request scopes.

7-Day Grace Threshold: Sectors refreshed within 7 days suffer 0 decay. Once unrun for >7 days, 15 defense points are deducted per full day elapsed past the grace threshold.

Neutral Transition: When defense reaches 0, the zone owner_id is set to None and defense_points to 0, making it unclaimed.

Live WebSocket Broadcast: Emits {"event": "territory_decay", "zone_id": id, "is_neutral": bool} to update active client maps.

7. ZoneCoach AI Engine with Medical Safety Guardrails

The conversational AI coach leverages Groq LLM inference (Llama 3.3 70B Versatile) paired with strict server-side guardrails. Unlike generic chatbots, the coach cannot recommend unsafe volume or pace.

- **Cardiovascular Ceiling:** If an athlete profile has documented hypertension, arrhythmia, or cardiac flags, the system forces all workouts into Zone 2 aerobic base, overrides tempo requests, and logs a medical constraint note.
- **10% ACWR Rule Enforcer:** Weekly progressive volume increases in adaptive training plans are capped at 10% maximum to preserve safe ACWR trajectories.
- **Audio Voice Transcription:** Athletes can speak running queries via Groq Whisper Large-v3 Turbo (<800ms transcription latency).

8. Enterprise Authentication, Tokens & Strava Webhooks

- **Dual JWT Token Architecture:** Short-lived 15-minute access tokens minimize breach impact; 7-day refresh tokens are cryptographically hashed (SHA-256) in the database with JTI rotation to detect and prevent replay attacks.
- **IDOR / BOLA Authorization:** Every activity, biometrics log, and route modification validates resource ownership against current_user.id or admin role.
- **Strava Push Webhooks:** Handles GET hub.challenge verification within 200ms and executes POST event payloads via FastAPI BackgroundTasks to fetch polyline streams, generate 40m PostGIS buffers, and capture territory asynchronously.

Behavioral & Technical Interview Q&A (STAR Method)

Structured talking points for senior full-stack and systems engineering interviews

Q: "Can you describe the most architecturally challenging feature you built in RunZone?"

[SITUATION] I needed to build a real-time multiplayer territory capture engine where runners claim GPS corridors on a live city map.

[TASK] Compute real-time 40m buffers around GPS coordinate tracks, detect overlaps against existing territories, calculate captured km², and update connected clients with <100ms latency.

[ACTION] I implemented PostGIS planar projections (ST_Transform to EPSG:3857) with ST_Buffer for millimeter-accurate corridors, used GiST spatial indexing for fast spatial collisions, and integrated a WebSocket broadcasting manager with FastAPI BackgroundTasks.

[RESULT] Reduced spatial overlap calculation from 1.2s to 75ms and supported concurrent territory captures with zero race conditions.

Q: "How did you handle the sector decay mechanics to ensure the game economy doesn't stagnate?"

[SITUATION] In territory conquest games, early adopters often lock down city sectors permanently, alienating new runners.

[TASK] Design an automated, reliable system that degrades inactive sectors over time and broadcasts updates in real time.

[ACTION] Created TerritoryDecayService with a pure mathematical decay function (-15 pts/day after 7 days grace). Wired it to an in-process AsyncIOScheduler in FastAPI lifespan that opens its own session and catches errors safely without crashing the server.

[RESULT] Achieved completely hands-free daily sweeps, neutral transitions for abandoned sectors, and automated live updates.

Q: "How did you prevent hallucinations or dangerous recommendations in your AI ZoneCoach?"

[SITUATION] LLMs can hallucinate aggressive running paces or ramp up mileage dangerously for beginners or athletes with medical conditions.

[TASK] Establish strict clinical guardrails before and after LLM inference to protect athlete health.

[ACTION] Built a dual-layer guardrail in Python and TypeScript. Injected user biometrics, resting HR, and medical flags into the prompt system, validated structured JSON outputs with Pydantic, and clamped volume progression to 10% ACWR limits.

[RESULT] 100% compliance with sports physiology guidelines; dangerous volume spikes are mathematically blocked before reaching the runner.

Architectural Trade-Offs, Scaling Plan & Candidate Pitch

Engineering trade-offs, system scaling strategies, and interview conclusion

9. Key Architectural Trade-Offs & Decisions

Architectural Trade-Off Analysis

PostGIS Server vs Client Turf.js: Decision: Server-authoritative PostGIS. Rationale: Client-side Turf.js calculations allow GPS spoofing and cannot maintain a shared multi-tenant spatial index across thousands of runners.

In-Process APScheduler vs Celery: Decision: APScheduler in FastAPI lifespan. Rationale: Avoided Celery/RabbitMQ broker overhead for a single-server deployment. The job runs in asyncio with its own DB session, keeping deployment lightweight.

WebSockets vs Short Polling: Decision: WebSocket event broadcasting. Rationale: Reduced map update traffic by 90% compared to 3-second HTTP polling, enabling instant notifications when a rival captures a nearby sector.

10. How Would You Scale RunZone to 500,000 Active Runners?

When interviewers ask: "How would you scale this system 100x?", here is the battle-tested roadmap:

- **1. Geospatial Caching & Vector Tile Server:** Pre-generate Mapbox Vector Tiles (.mvt) cached in Redis/Cloudflare CDN. Dynamic PostGIS queries only run for live edits, offloading 95% of map read traffic.
- **2. Asynchronous Worker Queue (Celery / BullMQ):** Extract Strava webhook processing and GPX polyline decoding into distributed Celery/Redis workers with auto-scaling Kubernetes pods.
- **3. Database Sharding & Read Replicas:** Implement PostgreSQL read replicas for leaderboard/analytics queries and partition territory_zones by geographic geohash / city bounds.
- **4. Edge Rate Limiting & Webhook Idempotency:** Place Cloudflare WAF in front of auth and webhook endpoints with distributed Redis token bucket rate limiting.

11. Summary of Interview Takeaways

Core Competencies Demonstrated in RunZone

Full-Stack Proficiency: FastAPI async microservices, SQLAlchemy 2.0 ORM, React 19, TypeScript strict mode, Tailwind CSS.

Geospatial Engineering: PostGIS 3.4, ST_Buffer, ST_Intersects, ST_Difference, GiST spatial indexing, WGS84 to Web Mercator.

Applied Data Science & Health: Dr. Tim Gabbett ACWR calculation, TRIMP exponential exertion models, HRV autonomic recovery scoring.

System Reliability: Token rotation, IDOR security, automated APScheduler sweeps, truth-in-UI honest states, unit test suites.

